

Roll No.

TEC-551

**B. TECH. (ECE ESR) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

SENSOR TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Describe the following terms in detail : (CO1)

(i) Calibration

(ii) Span

(iii) Full Scale output

OR

(b) Write in detail about hysteresis and non-linearity with suitable diagram. (CO1)

2. (a) Discuss about accuracy, saturation, and dead band in detail. (CO1)

OR

(b) Explain about repeatability, reliability, resolution, and uncertainty. (CO1)

P.T.O.

(2)

3. (a) Write a short note on the importance of electric charges, electric fields and potentials in terms sensors. (CO2)

(b) Explain the working of capacitive level sensor and proximity sensor. (CO2)

4. (a) Explain about piezoelectric effect and pyroelectric effect and its utilization in sensor fabrication. (CO2)

OR

(b) Explain about hall effect and thermoelectric effect with a suitable sensor example. (CO2)

5. (a) Explain about the working of an temperature sensor and infrared sensors in detail. (CO1, CO2)

OR

(b) Explain the quality of a good gas sensor in terms of its characteristics. (CO1, CO2)